PLSC 508 Replication and Extension: Network, Vertex Attributes, and Reproduction

Yasin Shafi

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Study and Repository

Study reproduced: Right et al. (2025a). Pushing Back or Backing Down? Evidence on Donor Responses to Restrictive NGO Legislation. *International Organization* 79(4): 759–786. doi:10.1017/S0020818325100957.

Repository (data, R code for every result in this document, and `README.txt`): `https://github.com/USERNAME/REPO`. The authors' original files are on Harvard Dataverse (Right et al., 2025b), doi:10.7910/DVN/FIINC8.

1 The Network

The original study is a dyadic panel analysis, not a network analysis. The extension treats the same data as a sequence of yearly two-mode (bipartite) networks, 2005–2019 (`code/02_build_network.R`).

- **Nodes, mode 1 (donors):** the 19 large OECD DAC donors in the original study.
- **Nodes, mode 2 (recipients):** aid-receiving countries in the original analysis sample, 105–109 per year. As in the paper, countries with population below 2 million in every year are excluded.
- **Ties:** donor i is tied to recipient j in year t if i's democracy-aid commitment to j in t is positive (OECD CRS commitments, the paper's democracy-sector coding). The commitment amount (USD) is stored as an edge attribute. A parallel economic-aid network uses economic-sector commitments.
- **Structure:** ties run only between modes. There are no donor–donor or recipient–recipient ties. The data are a balanced donor × recipient panel, so an absent tie is an observed zero, not missing data.

Donor–donor "co-funding" ties (two donors funding the same recipient) would come from a one-mode projection of this bipartite network. A projection discards information and mechanically creates clustering. Projections are therefore used only descriptively; the extension models the bipartite network directly. Table 1 reports size and density by year. Both networks are dense: about 60% of possible democracy-aid ties and 80% of possible economic-aid ties are present. The extension will test whether results depend on the "any positive commitment" tie definition by using higher thresholds.

Table 1: Democracy- and economic-aid networks: size and density by year

Year	Donors	Recipients	Dem. ties	Dem. density	Econ. ties	Econ. density	Recipients post-law
2005	19	109	940	0.454	1598	0.772	1
2006	19	109	941	0.454	1492	0.72	2
2007	19	109	1096	0.529	1713	0.827	4
2008	19	108	1160	0.565	1714	0.835	5
2009	19	108	1182	0.576	1714	0.835	7
2010	19	108	1242	0.605	1727	0.842	10
2011	19	107	1253	0.616	1740	0.856	14
2012	19	107	1229	0.605	1719	0.846	17
2013	19	107	1266	0.623	1697	0.835	21
2014	19	107	1192	0.586	1636	0.805	31
2015	19	107	1191	0.586	1649	0.811	39
2016	19	107	1195	0.588	1669	0.821	42
2017	19	107	1210	0.595	1645	0.809	47
2018	19	105	1317	0.66	1640	0.822	50
2019	19	105	1276	0.64	1628	0.816	52

Density = ties / (donors x recipients). Bipartite: only donor-recipient ties are possible.

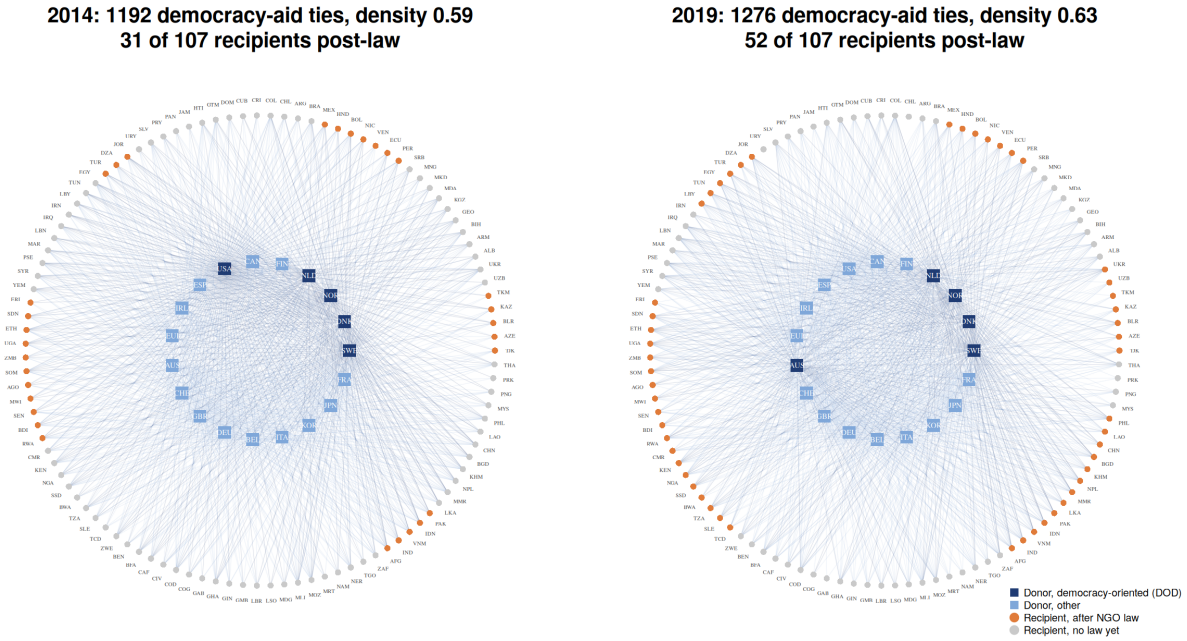


Figure 1: The democracy-aid network in 2014 and 2019, drawn with identical node positions. Squares are donors (inner ring, ordered by democracy-aid share; dark = democracy-oriented donor in that year, the paper's DOD measure). Circles are recipients (outer ring, grouped by region; orange = after its first restrictive NGO law). Edges are democracy-aid commitments. Source: `code/03_network_plot.R`.

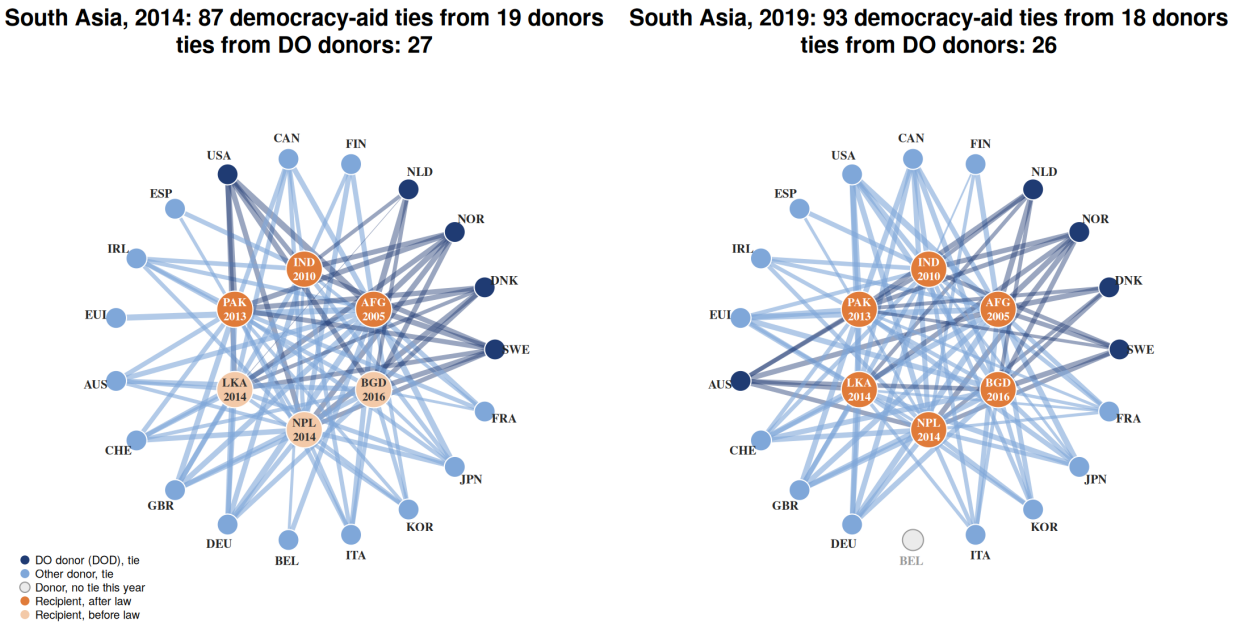


Figure 2: South Asian sub-network, 2014 and 2019. Recipients are labelled with the year of their first restrictive NGO law; edge width is the logged commitment amount, on one scale for both years. Bhutan and the Maldives are excluded by the population rule. Source: `code/03_network_plot.R`.

2 Vertex Attributes

Figures 3 and 4 show the distributions of the node attributes used in the extension (`code/04_vertex_attributes.R`). Donors differ sharply in how much of their aid goes to democracy and governance, which is the basis of the paper's democracy-orientation measures. Recipients span the full range of regime types; 54 of them pass a restrictive NGO law between 2005 and 2019, most of them after 2012.

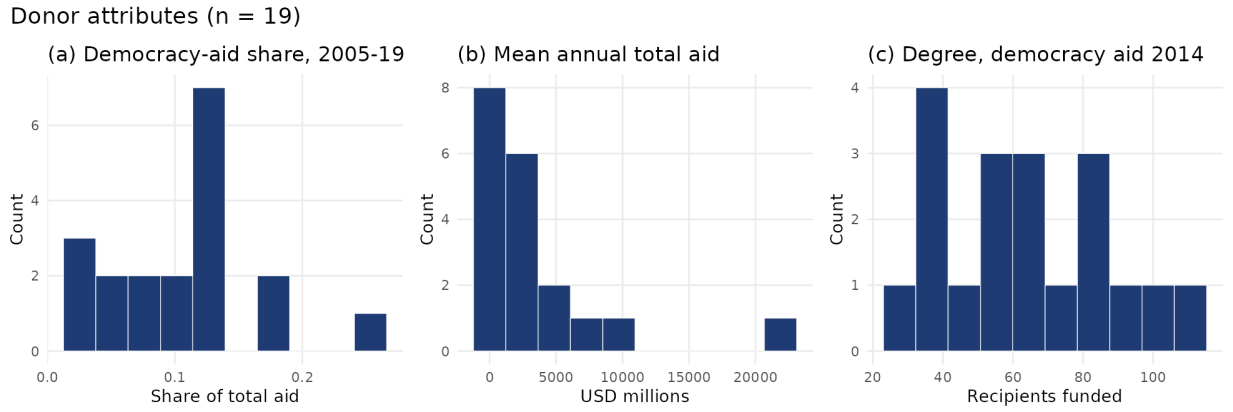


Figure 3: Donor attributes ($n = 19$): democracy-aid share of total aid (2005–2019), mean annual total commitments, and degree in the 2014 democracy-aid network.

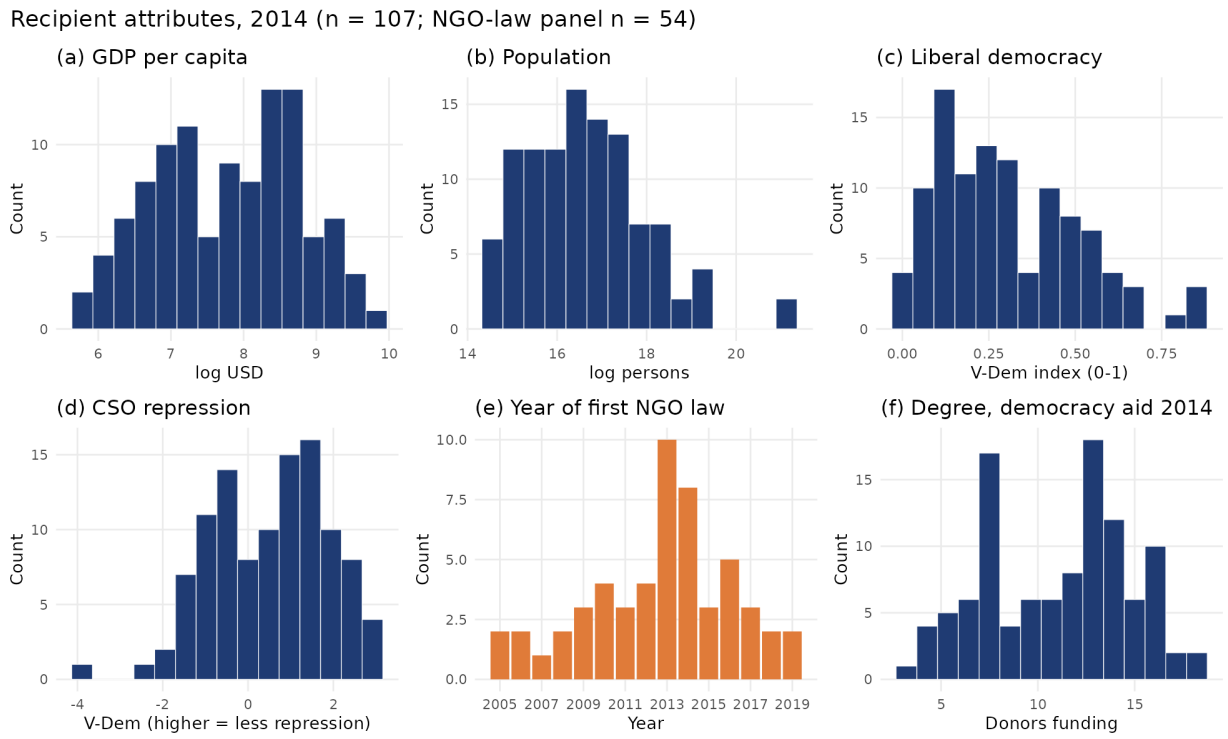


Figure 4: Recipient attributes in 2014: GDP per capita, population, V-Dem liberal democracy, V-Dem civil society repression, year of first restrictive NGO law (recipients that pass one), and degree in the 2014 democracy-aid network.

3 Reproduction of the Original Results

`code/01_replication.R` is the authors' replication script with relative paths and optional sections. Every two-way fixed-effects table it produces (Tables 1 and 2 and appendix Tables A3–A13 and the channel table, 14 in total) is identical, digit for digit, to the authors' published output. The only exception is a labelling fix in Table A10: the original code filled the year fixed-effects row for only three of the six columns, and the coefficients are unchanged. Table 2 and the table that follows it reproduce the paper's Tables 1 and 2. Following the authors' instructions, the generalized synthetic control models are not re-estimated (nine models with 1,000 bootstrap draws each). The figures are produced from the authors' saved estimates (`gsc_results.Rdata`, Dataverse). Figure 5 reproduces the paper's Figure 5.

Run `code/01_replication.R` with `data/gsc_results.Rdata` present to generate this figure.

Figure 5: Effect of law passage on democracy aid, by donor orientation (generalized synthetic control); reproduces the paper's Figure 5. Left: democracy-oriented donors. Right: other donors.

Table 2: Homogenous Response to NGO Law Enactment

	Dependent variable:		
	Total Aid	Democracy Aid	Economic Aid
	(1)	(2)	(3)
NGO Law Passage	0.057	−0.127	0.055
	(0.177)	(0.220)	(0.218)
Year FEs	✓	✓	✓
Dyad FEs	✓	✓	✓
Donor FEs	×	×	×
Number of Dyads	1064	1064	1064
Observations	14,402	14,400	14,401
R^2	0.661	0.591	0.669
Adjusted R^2	0.633	0.557	0.643
Note:			*p<0.1; **p<0.05; ***p<0.01

Table 3: Marginal Effect of Law Passage by Democracy Orientation

	Dependent variable:					
	Total Aid	Democracy Aid	Economic Aid	Total Aid	Democracy Aid	Economic Aid
	(1)	(2)	(3)	(4)	(5)	(6)
NGO Law Passage	0.076	−0.129	0.004	0.317*	0.307	0.220
	(0.389)	(0.548)	(0.454)	(0.165)	(0.226)	(0.207)
DO Relationship (DOR)	0.041	1.890***	−0.740***			
	(0.231)	(0.330)	(0.285)			
NGO Law * DOR	−0.404	−1.050***	−0.015			
	(0.258)	(0.274)	(0.282)			
DO Donor (DOD)				−0.372**	0.507**	−0.585***
				(0.158)	(0.198)	(0.182)
NGO Law * DOD				−0.772***	−1.485***	−0.486**
				(0.185)	(0.225)	(0.241)
Year FEs	✓	✓	✓	✓	✓	✓
Dyad FEs	×	×	×	✓	✓	✓
Donor FEs	✓	✓	✓	×	×	×
No. of Dyad FE	NA	NA	NA	1045	1045	1045
No. of Donor FE	19	19	19	NA	NA	NA
Observations	12,785	12,783	12,784	12,312	12,310	12,311
R^2	0.176	0.203	0.200	0.705	0.620	0.706
Adjusted R^2	0.174	0.201	0.198	0.678	0.584	0.678
Note:						*p<0.1; **p<0.05; ***p<0.01

A Reproduced Robustness Tables

All tables below are generated by `code/01_replication.R` and match the published appendix.

Table 4: Staggered DID: Inclusion of Never-Treated Dyads

	Dependent variable:		
	Total Aid	Democracy Aid	Economic Aid
	(1)	(2)	(3)
NGO Law Passage	0.130	0.260	0.064
	(0.144)	(0.226)	(0.176)
DOD	-0.609^{***}	0.138	-0.526^{***}
	(0.098)	(0.121)	(0.108)
NGO Law*DOD	-0.613^{***}	-1.236^{***}	-0.525^{**}
	(0.180)	(0.203)	(0.227)
Year FEs	✓	✓	✓
Dyad FEs	✓	✓	✓
Donor FEs	×	×	×
No. of Dyad FEs	2071	2071	2071
No. of Donor FEs	NA	NA	NA
Observations	24,377	24,373	24,376
R^2	0.714	0.629	0.720
Adjusted R^2	0.687	0.595	0.694
Note:			*p<0.1; **p<0.05; ***p<0.01

Table 5: Alternative Coding of Democracy Aid

	Dependent variable:					
	Total Aid	Democracy Aid	Economic Aid	Total Aid	Democracy Aid	Economic Aid
	(1)	(2)	(3)	(4)	(5)	(6)
NGO Law Passage	0.086	0.086	0.190	0.140	0.132	0.222
	(0.385)	(0.524)	(0.437)	(0.172)	(0.232)	(0.180)
DO Relationship (DOR)	0.038	1.445^{***}	-0.694^{**}			
	(0.266)	(0.350)	(0.333)			
NGO Law * DOR	-0.442^{*}	-0.897^{***}	-0.104			
	(0.254)	(0.261)	(0.299)			
DO Donor (DOD)				-0.522^{***}	-0.147	-0.673^{***}
				(0.131)	(0.152)	(0.132)
NGO Law * DOD				-0.099	-0.412^{*}	0.160
				(0.165)	(0.219)	(0.186)
Year FEs	✓	✓	✓	✓	✓	✓
Dyad FEs	×	×	×	✓	✓	✓
Donor FEs	✓	✓	✓	×	×	×
No. of Dyad FE	NA	NA	NA	1045	1045	1045
No. of Donor FE	19	19	19	NA	NA	NA
Observations	12,785	12,784	12,784	12,312	12,311	12,311
R^2	0.176	0.173	0.201	0.704	0.629	0.716
Adjusted R^2	0.174	0.171	0.199	0.676	0.594	0.689
Note:						*p<0.1; **p<0.05; ***p<0.01

Table 6: Continuous Measure of Democracy Orientation

	Dependent variable:					
	Total Aid	Democracy Aid	Economic Aid	Total Aid	Democracy Aid	Economic Aid
	(1)	(2)	(3)	(4)	(5)	(6)
NGO Law Passage	0.066	0.043	−0.030	0.656***	1.260***	0.546**
	(0.399)	(0.556)	(0.454)	(0.203)	(0.311)	(0.271)
DOR (continuous)	−0.026***	0.019*	−0.053***			
	(0.009)	(0.011)	(0.010)			
NGO Law * DOR (continuous)	−0.017**	−0.030***	−0.016**			
	(0.008)	(0.008)	(0.007)			
DOD (continuous)				−0.089***	−0.025	−0.088***
				(0.022)	(0.024)	(0.024)
NGO Law * DOD (continuous)				−0.048***	−0.119***	−0.040**
				(0.014)	(0.019)	(0.019)
Year FEs	✓	✓	✓	✓	✓	✓
Dyad FEs	×	×	×	✓	✓	✓
Donor FEs	✓	✓	✓	×	×	×
No. of Dyad FE	NA	NA	NA	1045	1045	1045
No. of Donor FE	19	19	19	NA	NA	NA
Observations	12,785	12,783	12,784	12,312	12,310	12,311
R^2	0.196	0.198	0.243	0.707	0.622	0.707
Adjusted R^2	0.194	0.196	0.241	0.679	0.586	0.679
Note:						*p<0.1; **p<0.05; ***p<0.01

Table 7: Dropping Dyads with USA

	Dependent variable:					
	Total Aid	Democracy Aid	Economic Aid	Total Aid	Democracy Aid	Economic Aid
	(1)	(2)	(3)	(4)	(5)	(6)
NGO Law Passage	0.058	−0.132	−0.018	0.318*	0.317	0.212
	(0.400)	(0.566)	(0.467)	(0.169)	(0.232)	(0.211)
DO Relationship (DOR)	0.103	1.942***	−0.694**			
	(0.247)	(0.350)	(0.302)			
NGO Law * DOR	−0.389	−1.091***	0.033			
	(0.270)	(0.288)	(0.296)			
DO Donor (DOD)				−0.447**	0.531**	−0.679***
				(0.191)	(0.228)	(0.220)
NGO Law * DOD				−0.808***	−1.590***	−0.486*
				(0.202)	(0.243)	(0.262)
Year FEs	✓	✓	✓	✓	✓	✓
Dyad FEs	×	×	×	✓	✓	✓
Donor FEs	✓	✓	✓	×	×	×
No. of Dyad FE	NA	NA	NA	990	990	990
No. of Donor FE	18	18	18	NA	NA	NA
Observations	12,090	12,088	12,089	11,664	11,662	11,663
R^2	0.140	0.153	0.168	0.692	0.596	0.694
Adjusted R^2	0.138	0.151	0.165	0.663	0.558	0.665
Note:						*p<0.1; **p<0.05; ***p<0.01

Table 8: Non-Zero Aid Flows Only

	Dependent variable:					
	Total Aid	Democracy Aid	Economic Aid	Total Aid	Democracy Aid	Economic Aid
	(1)	(2)	(3)	(4)	(5)	(6)
NGO Law Passage	0.034	−0.031	0.056	0.099	0.125	0.066
	(0.237)	(0.174)	(0.236)	(0.097)	(0.113)	(0.107)
DO Relationship (DOR)	−0.235*	0.392***	−0.332**			
	(0.127)	(0.105)	(0.132)			
NGO Law * DOR	−0.005	−0.200	0.025			
	(0.135)	(0.136)	(0.149)			
DO Donor (DOD)				−0.221***	0.051	−0.278***
				(0.066)	(0.089)	(0.067)
NGO Law * DOD				−0.257***	−0.306***	−0.192*
				(0.090)	(0.108)	(0.101)
Year FEs	✓	✓	✓	✓	✓	✓
Dyad FEs	×	×	×	✓	✓	✓
Donor FEs	✓	✓	✓	×	×	×
No. Dyad FEs	NA	NA	NA	1036	1036	1036
No. Donor FEs	19	19	19	NA	NA	NA
Observations	11,856	8,377	11,232	11,252	8,061	10,618
R^2	0.274	0.271	0.273	0.796	0.685	0.780
Adjusted R^2	0.272	0.268	0.271	0.775	0.641	0.756
Note:						*p<0.1; **p<0.05; ***p<0.01

Table 9: IHS-transformed Outcomes

	Dependent variable:					
	Total Aid (ihs)	Democracy Aid (ihs)	Economic Aid (ihs)	Total Aid (ihs)	Democracy Aid (ihs)	Economic Aid (ihs)
	(1)	(2)	(3)	(4)	(5)	(6)
NGO Law Passage	0.078	−0.138	0.012	0.330*	0.320	0.233
	(0.397)	(0.571)	(0.465)	(0.171)	(0.235)	(0.215)
DO Relationship (DOR)	0.053	1.978***	−0.754**			
	(0.238)	(0.344)	(0.296)			
NGO Law * DOR	−0.422	−1.098***	−0.020			
	(0.268)	(0.287)	(0.292)			
DO Donor (DOD)				−0.385**	0.529**	−0.610***
				(0.165)	(0.207)	(0.192)
NGO Law * DOD				−0.801***	−1.565***	−0.502*
				(0.194)	(0.235)	(0.253)
Year FEs	✓	✓	✓	✓	✓	✓
Dyad FEs	×	×	×	✓	✓	✓
Donor FEs	✓	✓	✓	×	×	×
No. of Dyad FE	NA	NA	NA	1045	1045	1045
No. of Donor FE	19	19	19	NA	NA	NA
Observations	12,785	12,785	12,785	12,312	12,312	12,312
R^2	0.172	0.200	0.198	0.699	0.615	0.700
Adjusted R^2	0.170	0.198	0.195	0.671	0.578	0.672
Note:						*p<0.1; **p<0.05; ***p<0.01

Table 10: Inclusion of Covariates

	Dependent variable:					
	Total Aid	Democracy Aid	Economic Aid	Total Aid	Democracy Aid	Economic Aid
	(1)	(2)	(3)	(4)	(5)	(6)
NGO Law Passage	0.087	0.037	−0.045	0.320*	0.282	0.147
	(0.292)	(0.417)	(0.310)	(0.169)	(0.220)	(0.193)
DO Relationship (DOR)	0.225	2.128***	−0.529*			
	(0.250)	(0.366)	(0.308)			
DO Donor (DOD)				−0.345**	0.455**	−0.601***
				(0.158)	(0.201)	(0.188)
Dyadic Trade	0.729***	1.079***	0.825***	0.095	0.073	0.117
	(0.107)	(0.130)	(0.119)	(0.086)	(0.112)	(0.083)
Recipient Population	−0.136	−0.558***	−0.151	−3.284	−5.161	−0.463
	(0.209)	(0.193)	(0.235)	(2.300)	(3.221)	(3.279)
Recipient GDP per capita	−2.063***	−2.334***	−2.578***	−1.820**	−1.892*	−1.153
	(0.251)	(0.285)	(0.285)	(0.754)	(1.038)	(0.927)
Democracy	0.285	0.250	1.332	0.022	1.539	0.009
	(1.650)	(1.926)	(1.971)	(1.975)	(1.999)	(2.679)
Human Rights	−0.420	−0.942***	−0.455	−0.389*	−0.693***	−0.388
	(0.274)	(0.333)	(0.338)	(0.213)	(0.254)	(0.252)
CSO Repression	0.264	0.390*	0.136	0.331*	0.535**	0.207
	(0.231)	(0.231)	(0.275)	(0.191)	(0.223)	(0.227)
Civil War	0.004	−0.107	−0.151	0.384***	0.369	0.508***
	(0.265)	(0.399)	(0.311)	(0.138)	(0.237)	(0.181)
Disaster Deaths	0.034	0.002	0.042	0.002	−0.022	0.005
	(0.028)	(0.040)	(0.031)	(0.013)	(0.016)	(0.013)
NGO Law * DOR	−0.538**	−1.241***	−0.266			
	(0.269)	(0.296)	(0.306)			
NGO Law * DOD				−0.805***	−1.452***	−0.514**
				(0.193)	(0.230)	(0.242)
Year FEs	✓	✓	✓	✓	✓	✓
Dyad FEs	×	×	×	✓	✓	✓
Donor FEs	✓	✓	✓	×	×	×
No. of Dyad FE	NA	NA	NA	954	954	954
No. of Donor FE	18	18	18	NA	NA	NA
Observations	11,551	11,549	11,550	11,001	10,999	11,000
R^2	0.327	0.314	0.352	0.720	0.645	0.723
Adjusted R^2	0.324	0.312	0.350	0.693	0.611	0.696

Note: *p<0.1; **p<0.05; ***p<0.01

Table 11: Controlling for Democratic Backsliding

	Dependent variable:					
	Total Aid	Democracy Aid	Economic Aid	Total Aid	Democracy Aid	Economic Aid
	(1)	(2)	(3)	(4)	(5)	(6)
NGO Law Passage	0.381	0.264	0.362	0.323*	0.338	0.202
	(0.343)	(0.484)	(0.402)	(0.163)	(0.231)	(0.203)
DO Relationship	0.056	1.910***	−0.724***			
	(0.228)	(0.329)	(0.280)			
DO Donor				−0.382**	0.506**	−0.598***
				(0.158)	(0.199)	(0.183)
Clean Elections	−1.509	−0.883	−1.582	−0.686	−0.049	−0.105
	(1.318)	(1.586)	(1.466)	(1.111)	(1.120)	(1.286)
Freedom of Expression	5.888***	7.154***	6.870***	2.315	4.507**	0.820
	(1.545)	(1.653)	(1.909)	(1.990)	(2.115)	(1.963)
Media Harassment	−0.850*	−1.148**	−1.072*	0.125	−0.013	0.185
	(0.485)	(0.472)	(0.588)	(0.202)	(0.277)	(0.241)
Attacks on Journalists	0.557***	0.706***	0.564**	0.142***	0.182**	0.191**
	(0.188)	(0.216)	(0.219)	(0.049)	(0.075)	(0.072)
Opposition Seat Share	−0.178	−0.773	−0.052	0.401	−0.602	0.647
	(0.945)	(1.059)	(1.112)	(0.365)	(0.545)	(0.464)
NGO Law * DOR	−0.400	−1.050***	−0.010			
	(0.259)	(0.275)	(0.283)			
NGO Law * DOD				−0.766***	−1.489***	−0.480*
				(0.186)	(0.225)	(0.242)
Year FEs	✓	✓	✓	✓	✓	✓
Dyad FEs	×	×	×	✓	✓	✓
Donor FEs	✓	✓	✓	×	×	×
No. of Dyad FE	NA	NA	NA	1045	1045	1045
No. of Donor FE	19	19	19	NA	NA	NA
Observations	12,766	12,764	12,765	12,293	12,291	12,292
R^2	0.217	0.232	0.238	0.709	0.624	0.708
Adjusted R^2	0.215	0.229	0.235	0.681	0.589	0.680
Note:						*p<0.1; **p<0.05; ***p<0.01

Table 12: Effect of NGO Laws on Democracy By Channel

	Dependent variable:							
	NGOs	Bypass	Recipient Gov't	Donor Gov't	NGOs	Bypass	Recipient Gov't	Donor Gov't
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
NGO Law Passage	−1.099***	−1.390***	0.299	−0.837**	0.033	0.004	0.288**	0.235
	(0.373)	(0.412)	(0.255)	(0.333)	(0.193)	(0.317)	(0.124)	(0.171)
Year FEs	✓	✓	✓	✓	✓	✓	✓	✓
Dyad FEs	✓	✓	✓	✓	✓	✓	✓	✓
No. Dyad FEs	246	246	246	246	674	674	674	674
Observations	2,454	2,453	2,452	2,453	6,725	6,726	6,725	6,726
R^2	0.584	0.563	0.433	0.552	0.660	0.553	0.548	0.570
Adjusted R^2	0.536	0.512	0.368	0.500	0.621	0.503	0.497	0.521
Note:								*p<0.1; **p<0.05; ***p<0.01

References

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